

Linear Regression

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1 Model Fitting

1.1 Model Assumption

Consider the general form a linear model with p variables (1 intercept and $p - 1$ slope variables).

$$y|x = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_{p-1} x_{p-1} + \epsilon \quad (1.1.1)$$

We assume this model is absolutely correct. Even if the model is not appropriate, this should be detected in the model examination process but not at this stage. An observation is a pair (y_i, x_i^T) where

$$x_i^T = (1, x_{i1}, \dots, x_{i,p-1})$$

for $i = 1, \dots, n$. We also hold these assumptions:

Assumption 1.1.1. *For the Multiple Linear Regression Model, we assume:*

1. $\epsilon_1, \dots, \epsilon_n \stackrel{i.i.d}{\sim} N(0, \sigma^2)$.
2. $y_i \sim N(x_i^T \beta, \sigma^2)$ for independent $i = 1, \dots, n$ such that $\mathbb{E}(y_i) = x_i^T \beta$, $\text{Var}(y_i) = \text{Var}(\epsilon_i) = \sigma^2$.

For simplicity, we can use the matrix form

$$Y|X \sim MVN(X\beta, \sigma^2 I) \quad (1.1.2)$$

for $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$ where Y_i is the i -th response and i -th row of X is the i -th observation,

$$Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad X = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1,p-1} \\ 1 & x_{21} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{n,p-1} \end{bmatrix}$$

The loss function is

$$L(X, Y; \beta) = \sum_{i=1}^n \underbrace{(y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_{p-1} x_{i,p-1})^2}_{\text{error } \epsilon_i} \quad (1.1.3)$$

or equivalently

$$L(X, Y; \beta) = \|Y - X\beta\|^2 := \|R\|^2 \quad (1.1.4)$$

where $\|\cdot\|$ is assumed to be the 2-norm. Same as in the SLR, we aim to find the optimized $\hat{\beta}$ that minimizes the total loss L .

1.2 Least Squares Estimate

The score function (gradient) of the loss function (1.1.4) is

$$S(X, Y; \beta) = \nabla_{\beta} L = -2X^T(Y - X\beta) \quad (1.2.1)$$

The second derivative (negative of information) is

$$I(X, Y; \beta) = \nabla_{\beta}^{(2)} L = 2X^T X^{1.1} \succ 0 \quad (1.2.2)$$

thus the loss function is convex. To find the derivatives, note that

$$\nabla_R L = \nabla_R R^T R = 2R$$

and

$$\nabla_{\beta} R = \nabla_{\beta} Y - X\beta = -X$$

therefore

$$\nabla_{\beta} L = \left(\frac{\partial R}{\partial \beta} \right)^T \left(\frac{\partial L}{\partial R} \right) = -2X^T(Y - X\beta)$$

Set to 0,

$$-2X^T(Y - X\beta) = 0$$

which implies that a minimizer $\hat{\beta}$ is a solution to

$$(X^T X)\hat{\beta} = X^T Y \quad (1.2.3)$$

Note that since the loss function (1.1.4) is convex, any of its stationary point is the global minimizer. That said any solution to (1.2.3) is an optimal solution. One possible choice could be the minimum-norm solution

$$\hat{\beta} = (X^T X)^{\dagger} X^T Y^{1.2} \quad (1.2.4)$$

and $\hat{\beta}$ has a unique solution

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad (1.2.5)$$

if and only if the columns of X are linearly independent^{1.3}. Note that the sum of squared residuals $\|R\|^2 = \|Y - X\beta\|^2$ is minimized such that for all $\beta \in \mathbb{R}^p$,

$$\|Y - X\hat{\beta}\|^2 \leq \|Y - X\beta\|^2 \quad (1.2.6)$$

1.3 Statistical Properties of LSE

When we have the estimated $\hat{\beta}$, we can find the fitted values

$$\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y \quad (1.3.1)$$

and fitted residuals

$$\hat{R} = Y - \hat{Y} \quad (1.3.2)$$

Let the *projection matrix* be denoted by

$$H = X(X^T X)^{-1} X^T \quad (1.3.3)$$

such that $\hat{Y} = HY$ and $\hat{R} = (I - H)Y$. As you can see, the projection matrix has useful properties.

^{1.1} $X^T X$ is positive definite for non-zero X .

^{1.2} $X^{\dagger}$ indicates the psuedo-inverse of X .

^{1.3} $X^T X$ is invertible iff X 's columns are linearly independent (assumed in our model). For better prediction power, we also require the angles between X 's column vectors to be large, not only linearly independent.

Property 1.3.1. *The projection matrix H is symmetric, idempotent, and positive semidefinite.*

It's obvious that it is symmetric, and note that

$$HH = X(X^T X)^{-1} X^T X (X^T X)^{-1} X^T = X(X^T X)^{-1} X^T = H$$

As the name suggests, H projects Y onto the column space of X such that $\hat{Y}$ is the projected^{1.4} vector in $\text{Col}(X)$. Normally, the system $X\beta = Y$ does not have an exact solution^{1.5}. To estimate a solution of $\hat{\beta}$, the LSE finds a vector $\hat{Y}$ in the column space of X that is the *closest* to Y , which is the projection $\hat{Y}$. There is a proof in linear algebra for why $\text{proj}_X Y$ is equivalent to the LSE solution. We first show that in $\text{Col}(X)$, $\text{proj}_X Y$ is the closest to Y . We have

$$\|Y - X\beta\|^2 = \|Y\|^2 - 2\langle Y, X\beta \rangle + \|X\beta\|^2$$

where $X\beta$ is any vector in $\text{Col}(X)$. Since $\text{proj}_X Y - X\beta \in \text{Col}(X) \perp \text{perp}_X Y$,

$$\begin{aligned} \|Y - X\beta\|^2 &= \|\text{proj}_X Y + \text{perp}_X Y - X\beta\|^2 \\ &= \underbrace{\|\text{perp}_X Y\|^2}_{\text{constant}} + \underbrace{\|\text{proj}_X Y - X\beta\|^2}_{\geq 0} + 2\langle \text{perp}_X Y, \text{proj}_X Y - X\beta \rangle \\ &\geq \|\text{perp}_X Y\|^2 \end{aligned}$$

The equality holds and thus $\|Y - X\beta\|^2$ is minimized when $\|\text{proj}_X Y - X\beta\|^2 = 0$, i.e. $X\hat{\beta} = \text{proj}_X Y$ as desired. Now we prove that the minimizer (1.2.5) implies $X\hat{\beta} = \text{proj}_X Y$. Substitute in the minimizer $\hat{\beta}$ and we find that

$$\langle \hat{Y}, \hat{R} \rangle = \hat{Y}^T \hat{R} = (HY)^T (I - H)Y = Y^T H^T Y - Y^T H^T H Y = 0 \quad (1.3.4)$$

and

$$\|Y - X\hat{\beta}\|^2 = \|Y\|^2 + \|X\hat{\beta}\|^2$$

such that $\hat{Y} \perp \hat{R}$, i.e. they are orthogonal and independent. Therefore, with $Y = \hat{Y} + \hat{R}$ and $\hat{Y} \perp \hat{R}$, we have

$$Y = \hat{Y} + \hat{R} = \text{proj}_X Y + \text{perp}_X Y \quad (1.3.5)$$

where Y is the target to be projected, $\hat{Y}$ is the projected vector in $\text{Col}(X)$, and $\hat{R}$ is the residual (difference) between them. $\square$

Note that (1.3.4) could also be written as

$$\langle \hat{Y}, \hat{R} \rangle = \sum_{i=0}^n \hat{r}_i \hat{y}_i = 0 \quad (1.3.6)$$

This explains that the fitted values and the residuals should be linearly independent. From the optimization (linear) constraint $S(X, Y; \beta) = -2X^T(Y - X\beta) = 0$, residuals are constrained to $X^T \hat{R} = 0$, namely

$$X^T \hat{R} = \sum_{i=1}^n x_{ij} \hat{r}_i = 0, \forall j = 0, \dots, p-1 \quad (1.3.7)$$

Since $x_{i0} = 1$ for all i , the first row of the above constraints is that all residuals sum to 0,

$$\sum_{i=1}^n \hat{r}_i = 0 \quad (1.3.8)$$

^{1.4}also called "fitted" in statistics

^{1.5}It has an exact solution $\hat{\beta} = X^{-1}Y$ only if X is invertible (saturated).

Derived from that, we have

$$\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i - \sum_{i=1}^n \hat{r}_i = \sum_{i=1}^n y_i - \sum_{i=1}^n \hat{r}_i = \sum_{i=1}^n y_i \quad (1.3.9)$$

such that y and $\hat{y}$ have the same total and mean. In general, with these p equalities in (1.3.7), the residuals are dependent and have $n - p$ degrees of freedom^{1.6} since X^T has p rows and n columns.

Now move on to the statistical implications of the LSE solution. Note that the loss function (1.1.3) is the negative of gaussian log likelihood function, so the LSE solution is indeed the MLE^{1.7} solution which follows a normal distribution.

Property 1.3.2 (Unbiasedness). *The least squares estimate $\hat{\beta} = (X^T X)^{-1} X^T Y$ is an unbiased estimator of β . Specifically, $\hat{\beta} \sim MVN(\beta, \sigma^2 (X^T X)^{-1})$.*

Recall that an estimator of some parameters is a random variable whose randomness comes from data. It a point estimate of the parameters whenever the data are observed and substituted in. To prove the unbiasedness, we need to show $\mathbb{E}(\hat{\beta}) = \mathbb{E}(\beta)$. We need this lemma:

Property 1.3.3. *If $Z \sim MVN(\mu, \Sigma)$, then $AZ \sim MVN(A\mu, A\Sigma A^T)$.*

Therefore

$$\begin{aligned} \mathbb{E}(\hat{\beta}) &= \mathbb{E}((X^T X)^{-1} X^T Y) \\ &= (X^T X)^{-1} X^T \mathbb{E}(Y) \\ &= (X^T X)^{-1} X^T X \beta \quad \text{recall } Y \sim MVN(X\beta, \sigma^2 I) \\ &= \beta \end{aligned}$$

also note that the covariance is

$$\begin{aligned} \mathbb{V}(\hat{\beta}) &= \mathbb{V}((X^T X)^{-1} X^T Y) \\ &= (X^T X)^{-1} X^T \mathbb{V}(Y) X (X^T X)^{-1} \\ &= (X^T X)^{-1} X^T (\sigma^2 I) X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1} \end{aligned}$$

By 1.3.3, we have

$$\hat{\beta} \sim MVN(\beta, \sigma^2 (X^T X)^{-1}) \quad (1.3.10)$$

and

$$\hat{Y} \sim MVN(X\beta, \sigma^2 H) \quad (1.3.11)$$

1.4 Variance Estimation

Recall the assumption in 1.1.1, we have that $\hat{r}_i = y_i - \hat{y}_i$ for $i = 1, \dots, n$ which should be a sequence of stationary variables following

$$\hat{R} \sim MVN(0, \sigma^2 (I - H)) \text{ and } \hat{r}_i \sim N(0, \sigma^2 (1 - h_{ii})), i = 1, \dots, n \quad (1.4.1)$$

where $\hat{R} = Y - \hat{Y} = (I - H)Y = (\hat{r}_1, \dots, \hat{r}_n)^T$. With $(I - H)^2 = (I - H)$, think of $Y, \hat{Y}$'s distributions and you will get this distribution. Since matrix X 's columns are well assumed to be linearly independent and $n \gg p$, the value h_{ii} should be very small and $\text{Var}(\hat{r}_i)$ is approximately σ^2 .

^{1.6}a.k.a. nullity of matrix X^T . The degree of freedom is the number of free parameters under certain constraints.

^{1.7}i.e. Maximum Likelihood Estimate

Now we want to estimate σ^2 based on the fitted model. By definition, $\sigma^2 = \mathbb{E}(y - x^T\beta|x)$. We don't know β , but we can use the estimator $\hat{\beta}$ to estimate $\sigma^2 = \mathbb{E}(y - x^T\hat{\beta}|x)$. We can estimate it via

$$s^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p} \quad (1.4.2)$$

The ANOVA states

Property 1.4.1 (ANOVA).

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{SST} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{SSE}$$

where:

$$SST := \text{Total Variance of } y$$

$SSR :=$ Sum Squares of Regression, the total variance of $\hat{y}$

SSE := Sum Squares of Error, measure of discrepancy between fitted values and original data

Proof.

$$\begin{aligned}\sum_{i=1}^n (y_i - \bar{y})^2 &= \sum_{i=1}^n (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y})\end{aligned}$$

In particular,

$$\begin{aligned} \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) &= \sum_{i=1}^n y_i \hat{y}_i - \sum_{i=1}^n \hat{y}_i^2 - \sum_{i=1}^n y_i \bar{y} + \sum_{i=1}^n \hat{y}_i \bar{y} \\ &= \sum_{i=1}^n (\hat{r}_i + \hat{y}_i) \hat{y}_i - \sum_{i=1}^n \hat{y}_i^2 - \underbrace{\bar{y} \sum_{i=1}^n y_i + \bar{y} \sum_{i=1}^n \hat{y}_i}_{=0 \text{ (1.3.9)}} \\ &= \underbrace{\sum_{i=1}^n \hat{r}_i \hat{y}_i}_{=0 \text{ (1.3.6)}} + \sum_{i=1}^n \hat{y}_i^2 - \sum_{i=1}^n \hat{y}_i^2 \\ &= 0 \end{aligned}$$

This proves that $SST = SSR + SSE$.

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Note that $SSR \perp SSE$ and

$$SSE/\sigma^2 \sim \chi^2(n-p)$$

An interpretation of this structure is

$$\underbrace{\|Y\|^2 - n\bar{y}^2}_{SST} = \underbrace{\|\hat{Y}\|^2 - n\bar{y}^2}_{SSR} + \underbrace{\|\hat{R}\|^2}_{SSE} \quad (1.4.3)$$

as of the orthogonality in (1.3.5). Let $MSE^{1.8} := SSE/(n-p)$ be denoted by s^2 as in (1.4.2). From (1.4.3), we have that:

^{1.8}Mean Squares of Error

Property 1.4.2 (Properties of Sum Squares). *The sum squares has properties:*

1. Only $\|R\|^2/\sigma^2$ (or SSE/σ^2) always follows a chi squared distribution of degree $n - p$. That said MSE is an unbiased estimator of σ^2 .
2. $SST \sim \chi^2(n - 1)$, $SSR \sim \chi^2(p - 1)$ only when under the hypothesis $\beta_1 = \dots = \beta_p = 0$, i.e. no linear effects exist.

Why it's the case? Note that $\mathbb{E}(\chi^2(k)) = k$, so $SSE/\sigma^2 \sim \chi^2(n - p) \implies \mathbb{E}(MSE) = \mathbb{E}(SSE)/(n - p) = \sigma^2$. Try rewriting SSE/σ^2 as

$$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{\hat{r}_i}{\sigma} \right)^2, \quad r_i/\sigma \stackrel{approx.}{\sim} N(0, 1)$$

$$\stackrel{approx.}{\sim} \chi^2(n - p), \quad \text{approx. sum of } n - p \text{ independent } \chi^2(1)$$

Recall (1.3.7), residuals have $n - p$ degrees of freedom. For SSR, SST , $\bar{y}$ is an unbiased predictor of y_i only when $\mathbb{E}(y_i) = \beta_0$ such that the model contains exactly one intercept parameter (constant model). Therefore, they are only valid chi squared distributions under the assumption $\beta_1 = \dots = \beta_p = 0$, especially for conducting hypothesis test on if linear effects exist.

1.5 Estimates and Confidence Intervals

By (1.3.10), we have the standardized $\hat{\beta}_j, j = 0, \dots, p - 1$ where

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{d_i}} \sim N(0, 1), \quad d_i = [(X^T X)^{-1}]_{ii} \quad (1.5.1)$$

The $100(1 - \alpha)\%$ confidence interval for β_j is

$$\left[\hat{\beta}_j \pm z_{1-\alpha/2} \sigma \sqrt{d_i} \right] \quad (1.5.2)$$

where $z_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$ ^{1.9}. Directly replace σ^2 with the estimated variance s^2 if the data sample is large, otherwise use t distribution as an alternate test statistics. To approach this, we need

Property 1.5.1 (t-Distribution). *The t-distribution is defined as*

$$T = \frac{Z}{\sqrt{\frac{W}{k}}} \sim t(k)$$

for independent $Z \sim N(0, 1)$, $W \sim \chi^2(k)$.

It will be asymptotically normal when k is large, so it will be indifferent to use t or normal as pivotal quantity when we have sufficient data. The test statistics and $100(1 - \alpha)\%$ confidence interval with t is

$$\frac{\hat{\beta}_j - \beta_j}{s \sqrt{d_i}} \sim t(n - p) \quad \left[\hat{\beta}_j \pm t_{1-\alpha/2}^{n-p} \sqrt{d_i} \right] \quad (1.5.3)$$

where $P(t(n - p) < t_{1-\alpha/2}^{n-p}) = 1 - \alpha/2$. For the proof, recall from (1.4.1) that $\frac{(n-p)s^2}{\sigma^2} \sim \chi^2(n - p)$, and consider

$$\frac{\hat{\beta}_j - \beta_j}{s \sqrt{d_i}} = \frac{(\hat{\beta}_j - \beta_j)/(\sigma \sqrt{d_i})}{\sqrt{\frac{(n-p)s^2/\sigma^2}{(n-p)}}}$$

^{1.9} $\Phi(z) = P(N_{0,1} < z)$

which follows $t(n-p)$ by 1.5.1.

Now we look at the residuals. By (1.4.1), for each $i = 1, \dots, n$,

$$\frac{\hat{r}_i}{\sigma\sqrt{1-h_{ii}}} \sim N(0, 1) \quad \left[\hat{r}_i \pm z_{1-\alpha/2}\sigma\sqrt{1-h_{ii}} \right] \quad (1.5.4)$$

and

$$\frac{\hat{r}_i}{s\sqrt{1-h_{ii}}} \sim t(n-p) \quad \left[\hat{r}_i \pm t_{1-\alpha/2}^{n-p}s\sqrt{1-h_{ii}} \right] \quad (1.5.5)$$

and they be used interchangeably when the data sample is large.

1.6 Prediction

With the fitted model, suppose we want to predict a new y_0 at a specific control variable x_0 where $y_0|x_0 \sim N(x_0^T\beta, \sigma^2)$. Note that y_0 is independent of the observed dataset $y_1, \dots, y_n$ such that

$$\mathbb{E}(y_0 y_i) = \mathbb{E}(y_0)\mathbb{E}(y_i), \forall i = 1, \dots, n \quad (1.6.1)$$

The mean response of $\hat{y}_0$ is

$$\hat{y}_0|x_0 = x_0^T \hat{\beta} \sim N(x_0^T \beta, \sigma^2 x_0^T (X^T X)^{-1} x_0) \quad (1.6.2)$$

The pivotal quantity for its confident interval is

$$\hat{y}_0 - y_0 \sim N(0, \sigma^2(1 + x_0^T (X^T X)^{-1} x_0)) \quad (1.6.3)$$

since y_0 is independent of $\hat{y}_0$. Therefore, the $100(1-\alpha)\%$ prediction interval for y_0 is

$$\left[\hat{y}_0 \pm z_{1-\alpha/2}\sigma\sqrt{1 + x_0^T (X^T X)^{-1} x_0} \right] \quad (1.6.4)$$

1.7 Special Case: Simple Linear Model

Suppose we have n data observations $(y_1, x_1), \dots, (y_n, x_n)$ and want to fit a simple linear model

$$y|x = \beta_0 + \beta_1 x + \epsilon \quad (1.7.1)$$

Based on these data, we want to find the coefficients $\hat{\beta}_0, \hat{\beta}_1$ that can effectively predict y_i for given x_i ,

$$y_i|x_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

We want to minimize the loss

$$L(X, Y; \beta) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \quad (1.7.2)$$

Take the derivative and set to 0,

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_{i=1}^n y_i - \beta_0 - \beta_1 x_i = 0 \quad \frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (1.7.3)$$

Solve,

$$\hat{\beta}_0 = \frac{1}{n} \sum_{i=1}^n y_i - \beta_1 \bar{x} = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\sum_{i=1}^n x_i y_i - x_i \hat{\beta}_0 - \hat{\beta}_1 x_i^2 = 0$$

Substitute in $\hat{\beta}_0$,

$$\begin{aligned}
\sum_{i=1}^n x_i y_i - x_i(\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i^2 &= 0 \\
\sum_{i=1}^n x_i y_i - x_i \bar{y} + \hat{\beta}_1 \bar{x} x_i - \hat{\beta}_1 x_i^2 &= 0 \\
\sum_{i=1}^n x_i (y_i - \bar{y}) + \hat{\beta}_1 x_i (\bar{x} - x_i) &= 0 \\
\hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x}) &= \sum_{i=1}^n x_i (y_i - \bar{y}) \\
\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n x_i (x_i - \bar{x})} \quad \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}
\end{aligned}$$

Note that

$$S_{xy} := \sum_{i=1}^n x_i (y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad S_{xx} := \sum_{i=1}^n x_i (x_i - \bar{x}) = \sum_{i=1}^n (x_i - \bar{x})^2 \quad (1.7.4)$$

So we can rephrase them to be

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (1.7.5)$$

The fitted values are thus

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \quad (1.7.6)$$

with residuals

$$\hat{r}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \quad (1.7.7)$$

Note that due to the optimization constraints in (1.7.3), the residuals are constrained to

$$\sum_{i=1}^n \hat{r}_i = 0 \quad \sum_{i=1}^n x_i \hat{r}_i = 0 \quad (1.7.8)$$

2 Model Selection and Examination

2.1 Examination Techniques

2.2 Residual Diagnostics

Recall that the fitted residuals are

$$\hat{R} = Y - \hat{Y} = Y - HY = (I - H)Y$$

Note that $I - H$ is also idempotent. By 1.3.3,

$$\hat{R} \sim MVN(0, \sigma^2(I - H)) \quad (2.2.1)$$

So the standardized residual is

$$\frac{r_i}{\sigma\sqrt{1 - h_{ii}}} \sim N(0, 1) \quad (2.2.2)$$

as well as the t distribution

$$\frac{r_i}{s\sqrt{1 - h_{ii}}} \sim t(n - p) \quad (2.2.3)$$